



Group 4
Data
Science

Nairobi Flood Guard AI

Predicting Floods. Protecting
Lives. Keeping Nairobi Moving.

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2026
Capstone
Project



A Crisis That Keeps Repeating

UNDERSTANDING WHY KENYA KEEPS FACING THE SAME DISASTER



2026

FLOODS
RETURN AGAIN

- April 2024: floods devastated Kenya — hundreds killed, thousands displaced, roads paralysed
- Nairobi's matatus were stranded, communities cut off, emergency services overwhelmed
- The same story repeated in 2026
- The root problem: flood response in Kenya is reactive — help arrives after the damage is done
- **We asked: what if we could see it coming?**



01



NAIROBI CITY COUNTY & NDMU

- Better flood planning and response
- Faster identification of high-risk areas
- Smarter resource allocation

02



MATATU OPERATORS & SACCOS

- Safer alternative routes during floods
- Reduced delays and transport disruptions
- Improved commuter safety

03



KENYA RED CROSS & RESPONDERS

- Faster emergency deployment
- Improved rescue coordination
- Better targeting of affected areas

04



EVERYDAY NAIROBIANS

- Earlier flood warnings
- Safer travel during floods
- Reduced exposure to flood risks

Who Stands to Benefit

Nairobi Flood Guard AI serves multiple stakeholders across the city, from government agencies coordinating emergency response to everyday commuters navigating flooded streets.

Our Answer: Two Tools, One System

2026

NAIROBI FLOOD
GUARD AI

**TWO INTEGRATED TOOLS WORKING TOGETHER
TO SHIFT FLOOD RESPONSE FROM REACTIVE TO
PREDICTIVE**



Flood Risk Map

- Predicts high-risk flood areas
- Generates ward-level risk scores

Safe Route Recommender

- Detects dangerous flooded roads
- Suggests safer alternative routes

Outcome

From reactive flood response → predictive flood prevention

What the System Looks At

DATA INPUTS FEEDING THE FLOOD PREDICTION SYSTEM

To predict flood risk in Kenya's 1,450 wards, the system analyzes:

Terrain Data:

- Elevation levels
- Slope steepness
- Water drainage patterns

Rainfall Data:

- Total rainfall amount
- Worst single-day rainfall
- Rainfall over the week prior to flooding

Each ward is assigned a flood risk score based on these factors.



The Insight That Changed Everything

Terrain Over Rainfall

- Flooding is driven by elevation and drainage patterns, not rainfall amounts.

Dry Counties Facing Flooding

- Turkana and Garissa, known as some of Kenya's driest regions, are now among the most affected by flooding.

#1 Predictor

- Elevation emerged as the strongest predictor of flood risk across all wards.

21%

OF WARDS FLOODED IN
APRIL 2024

1,450 WARDS ANALYSED

- Analyzed flood patterns across all Kenyan wards
- Found that flooding is driven more by terrain and water drainage than rainfall alone
- This insight reshaped our flood prediction model and risk analysis approach

01

THE SURPRISING FINDING

- Traditional models focus mainly on rainfall intensity

02

- Our analysis showed terrain and water flow are stronger flood drivers
- Low-lying areas flood heavily as water drains from higher elevations into basin regions

WHY THIS MATTERS

- Shifted flood prediction from rainfall to terrain-based analysis
- Used elevation, slope, and drainage patterns to predict flood risk
- Enabled proactive planning before flooding occurs

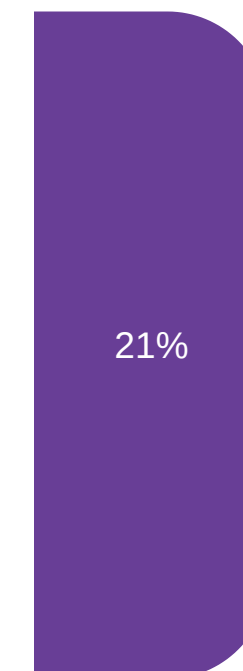
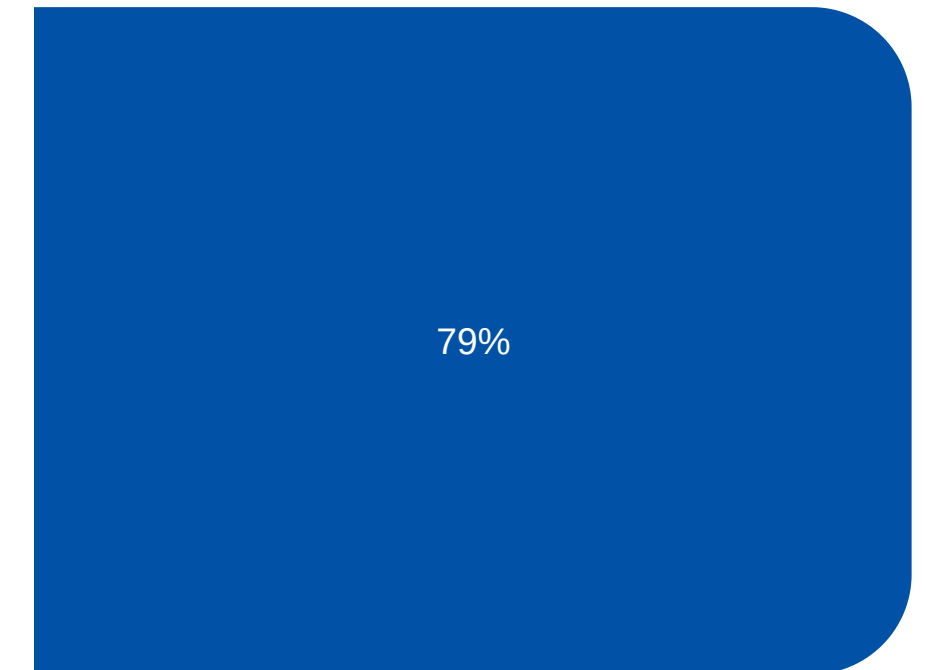
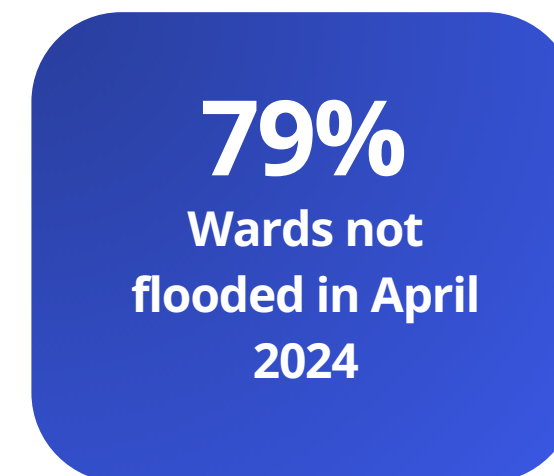
03

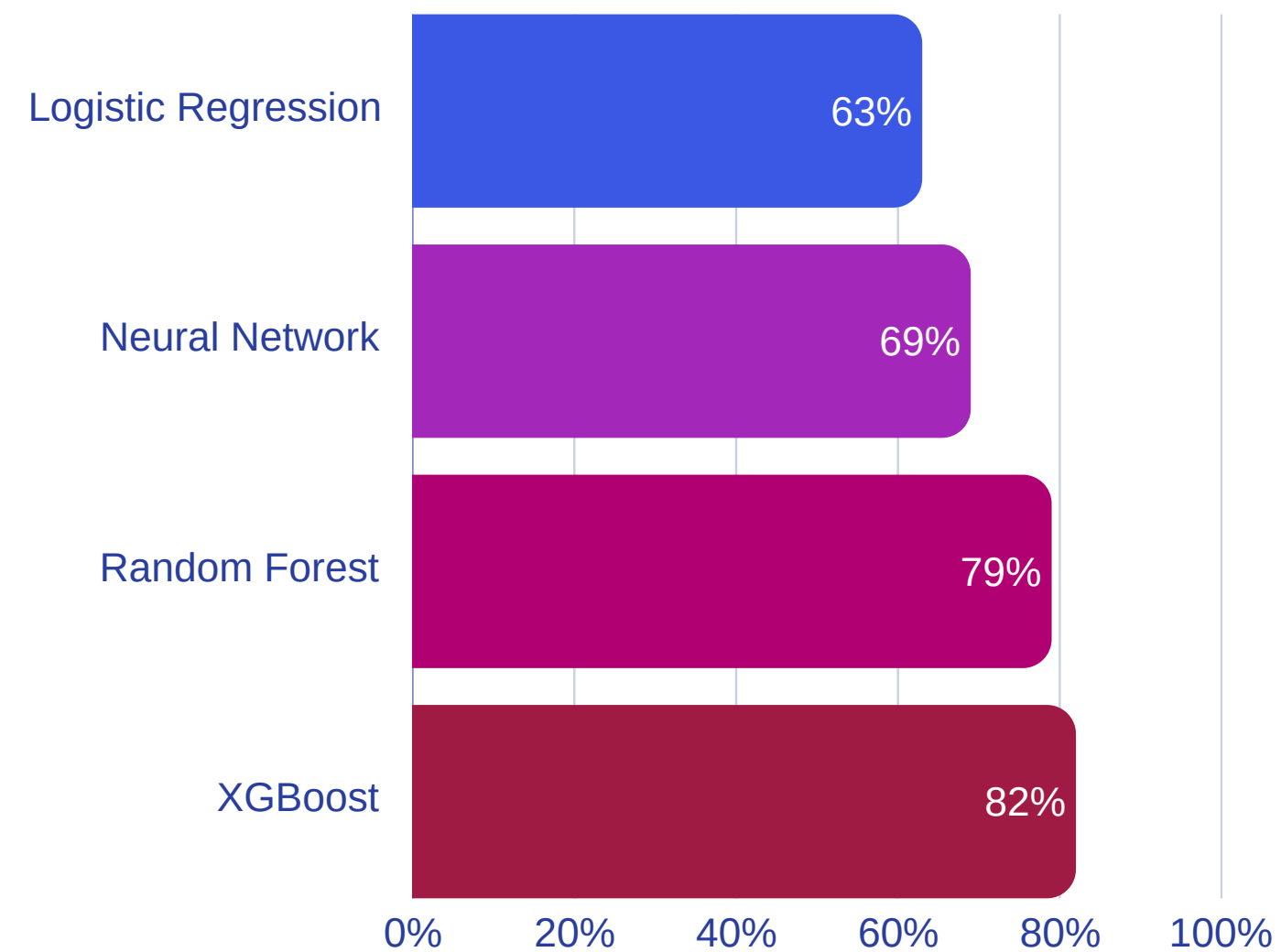
The Challenge: Most Wards Don't Flood

UNDERSTANDING THE DATA IMBALANCE THAT SHAPED OUR MODELING APPROACH



- Only 21% of Kenya's wards experienced flooding, making the dataset highly imbalanced
- A model predicting "no flood" everywhere could still achieve high accuracy while missing real flood events
- We prioritized **recall** to detect as many actual floods as possible, even at the cost of some false alarms, because missing floods poses greater risks to lives and infrastructure





82%

Recall score
achieved by
XGBoost model

Why We Chose XGBoost

WE EVALUATED 4 MODELS. XGBOOST WON ON THE METRIC THAT MATTERS MOST — RECALL.



- The model focuses on recall since missing floods is more critical than false alarms
- Four models were tested:
 - Logistic Regression (63%)
 - Neural Network (69%)
 - Random Forest (79%)
 - XGBoost performed best with 82% recall
- XGBoost identifies the most at-risk wards accurately
- A ward is classified as high-risk when flood probability exceeds 45%, a threshold that can be adjusted based on risk needs

01

Githunguri to Cabanas

34% flood risk reduction with only +13.7 minutes extra travel time.

02

Utawala to Kencom

28% flood risk reduction with +8 minutes extra travel time.

03

By Pass to Cabanas

25% flood risk reduction with +8.5 minutes extra travel time.

04

Quickmart to Muthurwa

26% flood risk reduction with +12.9 minutes extra travel time.

Keeping Nairobi's Matatus Moving Safely

HOW FLOOD RISK SCORES TRANSFORM MATATU ROUTING DECISIONS



- XGBoost flood-risk scores were integrated into Nairobi's road network
- Roads in high-risk wards were avoided and safer alternative routes recommended
- Rerouted matatu paths reduced flood exposure by up to 34%, improving commuter safety
- The system prioritizes safer travel, even when alternative routes increase travel time

Ready to Deploy Today

PRODUCTION-READY INFRASTRUCTURE —
NO NEW HARDWARE NEEDED



01

STREAMLIT WEB APP

Interactive map showing ward flood risk, affected stops, and rerouted paths accessible from any browser

02

GTFS-RT FEED

Rerouting decisions plug directly into Google Maps and Transit App, zero changes for operators

03

TUNABLE THRESHOLDS

Flood sensitivity and road penalties adjust in real time to match any event's severity

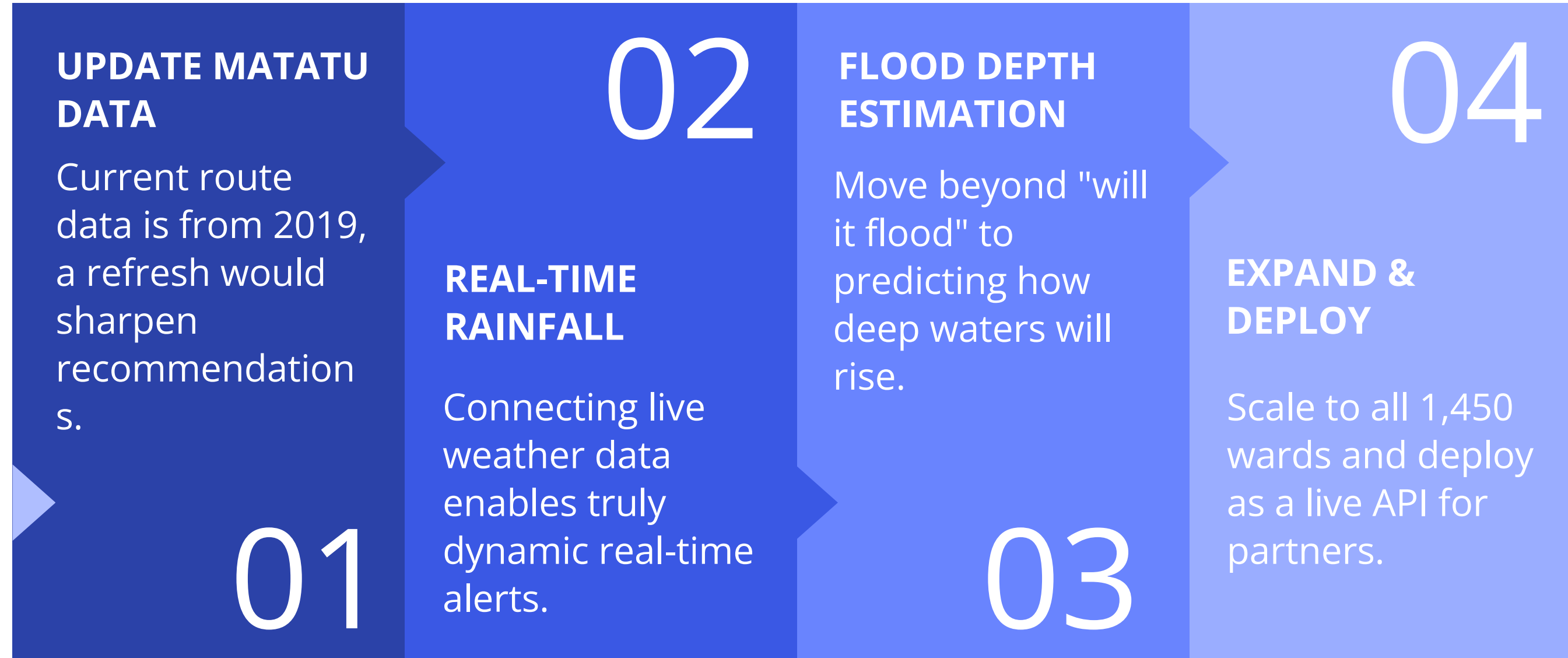
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INSTANT DEPLOYMENT

No new hardware required, runs on existing production ready infrastructure

The Road Ahead

**FIVE PRIORITIES TO TAKE NAIROBI FLOOD
GUARD FURTHER**



What We're Asking For

12

PARTNERSHIP & SUPPORT REQUESTS ↗

- » A pilot partnership with Nairobi City County, NDMU, or Kenya Red Cross to deploy and validate the system.
- » Access to real-time rainfall and road condition data to power the next version of flood predictions.
- » Support to update the 2019 GTFS matatu feed with current route data for accurate recommendations.
- » This system is built, tested, and ready — we just need the right partners to take it live.



Thank You!

Floods don't
wait.
Neither should
we.

Github:

https://github.com/T31SHA/Nairobi_Flood_Guard-AI



Group 4

**Nairobi
Flood Guard
AI**

Questions?

We're ready.